Spring 2023 - STAT2056 Final Exam – 14/6/2023 – 13:00 – 14:15

Question 1 – (30 pts)

A digital communication system transmits one of the two possible values, i.e., 0 and 6. A channel adds Gaussian noise with zero mean and variance 9 which may cause the decoder to make decision errors. At the receiver side, if the received sample is greater than 2, we decide symbol 6, otherwise we decide symbol 0

(a) (15 pts) If the probabilities for the symbols being transmitted are P(6) =1/4 and P(0)=3/4 find the probability of error.
(b) (5 pts) **If an error has been observed,** what is the probability that we have transmitted a symbol 6 ?.
(c) (10 pts) What is the probability that at the receiver we decide symbol 6?

Question 2 - (40 pts)

From a bag of fruit containing 3 oranges, 1 apple, and 2 bananas, a random sample of 2 pieces of fruit is selected. If X is the number of **oranges** and Y is the number of **apples** in the sample,

a. Find the joint probability distribution of X and Y. *(10 pts)*

b. Are X and Y independent? *(5 pts)*

c. Let A is the region that is given by $\{x + y \leq 1\}$. Find $P(A)$ *(5 pts)*

d. Find $E(X|Y=0)$. *(5 pts)*

e. Let $Z = X - Y$. Find the mean and variance of Z. *(15 pts)*

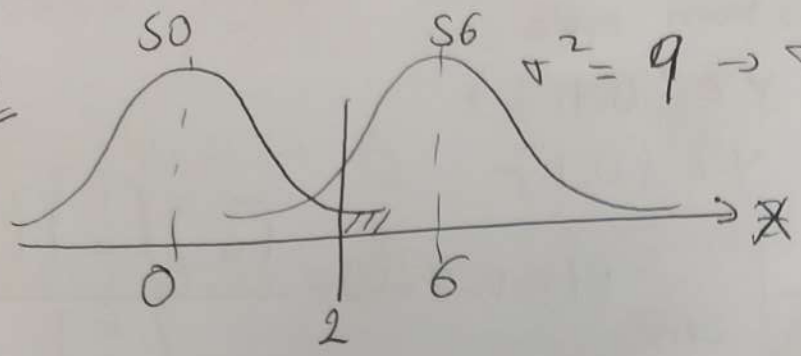
Question 3 - (30 pts)

The daily accumulated rainfall around Black Sea region during May is assumed to be normally distributed with a mean of 20 millimeters and a standard deviation of 5 millimeters: Suppose the rainfall amount is monitored for a month and the sample mean of 30 days $\overline{X}$ is computed. Assume that the sample mean can be measured to any degree of accuracy, and rainfall in each day is independent to other days observed in a month.

a. Determine the mean and standard deviation of the sample mean $\overline{X}$; (5 pts)
b. Find the probability that the sample mean is between 18 and 23 millimeters (10 pts)
c. Now suppose that the daily accumulated rainfall is **rounded to the nearest integer**. Approximate the probability that sample mean is equal to 20 millimeters. (15 pts)

STAT 2056 Final Exam Solutions

Q1



(a) $P_e = P(X>2 \mid S0)P(S0) + P(X<2 \mid S6)P(S6)$

$= \left[1 - \Phi\left(\frac{2-0}{3}\right)\right] \times \frac{3}{4} + \Phi\left(\frac{2-6}{3}\right) \times \frac{1}{4}$

$= \left[1 - \Phi\left(\frac{2}{3}\right)\right] \times \frac{3}{4} + \left[1 - \Phi\left(\frac{4}{3}\right)\right] \times \frac{1}{4}$

$= 0.2525 \times \frac{3}{4} + 0.0912 \times \frac{1}{4} = 0.2122$

(b) ~~$P(X=6 \mid E) = \frac{P(E \wedge X=6)}{P(E)} = P(X=6 \mid E)$~~

$P(S6 \mid E) = \frac{P(S6 \wedge E)}{P(E)} = \frac{P(E \mid S6)P(S6)}{P(E)}$

$= \frac{P(x<2 \mid S6)P(S6)}{P(E)} = \frac{0.0228}{0.2122} = 0.1075$

(c) $P(S6) = P(S6 \mid S0)P(S0) + P(S6 \mid S6)P(S6)$

$= P(X>2 \mid S0)P(S0) + P(X>2 \mid S6)P(S6)$

$= \left[1 - \Phi\left(\frac{2-0}{3}\right)\right] \times \frac{3}{4} + \left[1 - \Phi\left(\frac{2-6}{3}\right)\right] \times \frac{1}{4}$

$= 0.4166$

Q2 3 orange
1 apple
2 banana

select 2

X → Num. ~~apples~~ oranges
Y → Num. apple
$X \in \{0,1,2\}$
$Y \in \{0,1\}$

X \ Y	0	1	
0	1/15	2/15	3/15
1	6/15	3/15	9/15
2	3/15	0	3/15
	10/15	5/15	

$P(X=0,Y=0) = \frac{\binom{3}{0}\binom{1}{0}\binom{2}{2}}{\binom{6}{2}} = \frac{1}{15}$

$P(X=0,Y=1) = \frac{\binom{3}{0}\binom{1}{1}\binom{2}{1}}{\binom{6}{2}} = 2/15$

$P(X=1,Y=0) = \frac{\binom{3}{1}\binom{1}{0}\binom{2}{1}}{\binom{6}{2}} = \frac{6}{15}$

$P(X=1,Y=1) = \frac{\binom{3}{1}\binom{1}{1}\binom{2}{0}}{\binom{6}{2}} = \frac{3}{15}$

$P(X=2,Y=0) = \frac{\binom{3}{2}\binom{1}{0}\binom{2}{0}}{\binom{6}{2}} = \frac{3}{15}$

$P(X=2,Y=1) = 0$

(b) $P(X=1,Y=0) = P(X=1)\,P(Y=0)$

$\frac{6}{15} \neq \frac{9}{15} \times \frac{5}{15}$ → X and Y not independent.

(c) $\{X+Y \leq 1\}$

$P(X=0,Y=0) + P(X=0,Y=1) + P(X=1,Y=0)$

$= \frac{1}{15} + \frac{2}{15} + \frac{6}{15} = \frac{9}{15} = \frac{3}{5} = 0.6$

(d) $E[X|Y=0] = \sum_{x=0}^{2} x P(X=x|Y=0)$

$P(X=0|Y=0) = \frac{1}{10}$

$P(X=1|Y=0) = \frac{6}{10}$

$P(X=2|Y=0) = \frac{3}{10}$

$= 0\times\frac{1}{10} + 1\times\frac{6}{10} + 2\times\frac{3}{10}$

$= \frac{12}{10} = 1.2$ //

$E[X] = 0\times\frac{3}{15} + 1\times\frac{9}{15} + 2\times\frac{3}{15}$

$= 1$

(e) $Z = X - Y$

$E[Z] = E[X] - E[Y]$

$= 1 - \frac{1}{3} = \frac{2}{3}$

$E[Y] = 0\times\frac{10}{15} + 1\times\frac{5}{15} = \frac{1}{3}$

$Var(Z) = Var(X) + Var(Y) - 2\,Cov(X,Y)$

$Var(X) = E[X^2] - E^2[X] = 1.4 - 1 = 0.4$ //

$E[X^2] = 0^2\times\frac{3}{15} + 1^2\times\frac{9}{15} + 2^2\times\frac{3}{15} = \frac{9}{15} + \frac{12}{15} = \frac{21}{15} = 1.4$

$Var(Y) = E[Y^2] - E^2[Y] = \frac{1}{3} - \frac{1}{9} = p(1-p) = \frac{1}{3}\times\frac{2}{3} = \frac{2}{9}$

$E[Y^2] = 0^2\times\frac{10}{15} + 1^2\times\frac{5}{15} = \frac{1}{3}$

$Cov(X,Y) = E[XY] - E[X]E[Y] = \frac{3}{15} - 1\times\frac{1}{3} = \frac{1}{5} - \frac{1}{3}$

$= \frac{3}{15} - \frac{5}{15} = -\frac{2}{15}$

$E[XY] = 0.0\,P(X=0,Y=0) + 0\times1\,P(X=0,Y=1)$
$+ 1\times0\,P(X=1,Y=0) + 1\times1\,P(X=1,Y=1)$
$+ 2\times0\,P(X=2,Y=0) + 2\times1\,P(X=2,Y=1) = \frac{3}{15}$

3/15

$Var(Z) = 0.4 + 0.22 - 2\left(-\frac{2}{15}\right) = 0.4 + 0.22 + 0.27$

$Var(Z) = 0.89$

Q3 $\mu = 20ml$, $\sigma = 5ml$ $\quad \bar{X} = \frac{X_1 + X_2 + \cdots + X_{30}}{30}$

$E[\bar{X}] = \mu = 20ml$ $\quad Std(\bar{X}) = \sqrt{\frac{25}{30}} = \sqrt{\frac{5}{6}} = 0.9129$

$Var(\bar{X}) = \frac{\sigma^2}{n} = \frac{25}{30}$

$P(18 < \bar{X} < 23) = P(\bar{X} < 23) - P(\bar{X} < 18)$

$= \Phi\left(\frac{23-20}{\sqrt{\frac{25}{30}}}\right) - \Phi\left(\frac{18-20}{\sqrt{\frac{25}{30}}}\right)$

$= \Phi\left(\frac{3}{0.9129}\right) - \Phi\left(-\frac{2}{0.9128}\right)$

$= 0.9995 - 0.0142 = 0.9853$

$P(\bar{X} = 20) = P(19.5 < \bar{X} < 20.5)$

$= \Phi\left(\frac{20.5-20}{0.9129}\right) - \Phi\left(\frac{19.5-20}{0.9129}\right)$

$= 0.7081 - 0.2919 = 0.4162$